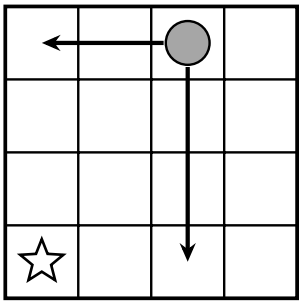
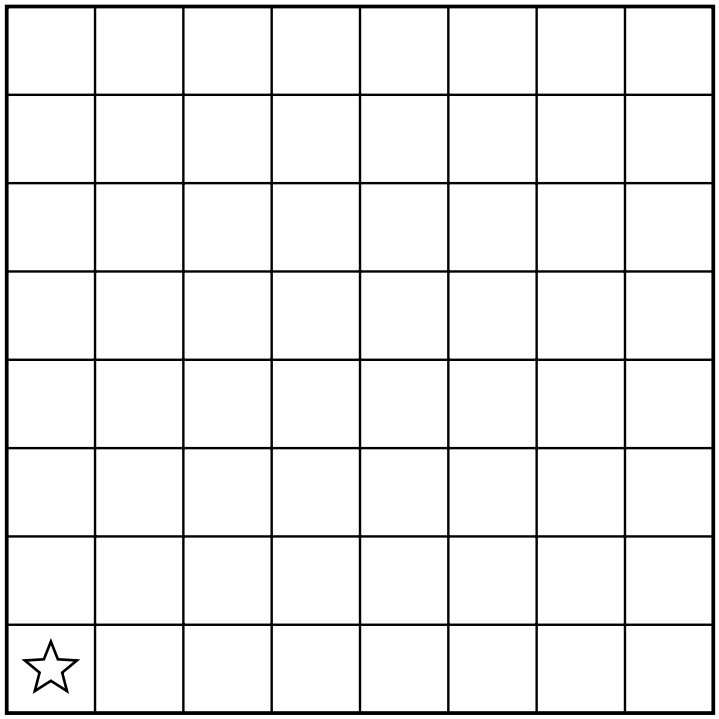
Players take turns. On a board, a move slides the token straight left or straight down, toward the star's side, one or more squares. Whoever moves the token onto the star wins.



In a game with piles of counters, a move takes one or more counters from one pile. Whoever takes the last counter wins.

Problem 1: Color every square of this 8-by-8 board except the star: green if the first player can always win when the token starts there, red if the second player can always win. Explain why the second player can always win from a red square.



Problem 3: Play many games with three piles from each start. Which player can always win?

1, 1, 1	_____	1, 1, 2	_____
1, 2, 3	_____	2, 2, 5	_____
1, 3, 4	_____	1, 4, 5	_____

Problem 4: Find all the starts with three piles, each with 1 to 7 counters, where the second player can always win. The order of the piles does not matter.

Problem 5: The second player can always win from each of these starts.

1, 2, 3 1, 4, 5 2, 4, 6 3, 5, 6 2, 5, 7

The first player can always win from each of these starts.

1, 2, 4 2, 3, 5 2, 4, 7 3, 4, 6

Make these starts with counters, and split every pile into stacks of 1, 2, 4, 8, 16 and so on, with no two stacks in one pile the same size. Find a rule about the stacks that tells which player can always win.

Problem 6: Which player can always win from each of these starts? When it is the first player, find a first move that lets the first player always win.

3, 5, 7 _____

6, 10, 12 _____

13, 9, 7 _____

11, 14, 21 _____